

Program: Diplomain Communication and Computer Networking	
Course Code: 3327	Course Title: Object Oriented Programming Lab using JAVA
Semester: 3	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To learn and implement object-oriented features such as encapsulation, inheritance and polymorphism along with error-handling techniques using Java.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	CognitiveLevel
CO1	Implement basic elements of JAVA	10	Applying
CO2	Implement Inheritance	8	Applying
CO3	Implement Exception handling	9	Applying
CO4	Implement more features Interfaces, Packages	10	Applying
	SeriesTest	4	
	OpenEndedExperiments/microprojects	4	

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			2			
CO2	3	3			3		
CO3	3			2			
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement basic elements of JAVA		
M1.01	Implement simple programs using OOP concepts (class, object, constructor, member functions).	2	Applying
M1.02	Implement a program having concepts of decision control structures and looping control structures	2	Applying
M1.03	Implement a program using method overloading	2	Applying
M1.04	Use Implement a program using command line arguments in java.	2	Applying
M1.05	Implement programs based on I/O operations using I/O classes.	2	Applying
CO2	Implement Inheritance		
M2.01	To implement program using Inheritance	4	Applying
M2.01	Implement programs based on Multiple Inheritance to demonstrate the use of interface.	4	
	Series Test-I	2	
CO3	Implement Exception Handling		
M3.01	Implement programs based on Exception handling.	9	Applying
CO4	Implement more features Interfaces, Packages		
M4.01	Implement interfaces	6	Applying
M4.02	Implement packages	4	
	Series Test- II	2	
	Open ended experiment/microprojects	4	

Note: Experiments shall be conducted such that all COs are attained.
 Minimum of 6 experiments (excluding open ended experiments / micro projects) shall be performed.
 Compulsory for CIA
 The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text/Reference

T/R	Book Title / Author
T1	Herbert Schildt and Dale Skrien, "Java Fundamentals – A comprehensive Introduction", McGraw Hill, 1st Edition, 2013.
T2	Herbert Schildt, "Java the complete reference", McGraw Hill, Osborne, 7th Edition, 2011
T3	T.Budd, "Understanding Object- Oriented Programming with Java", Pearson Education, Updated Edition (New Java 2 Coverage), 1999
R1	P.J.Dietel and H.M.Dietel, "Java How to program", Prentice Hall, 6th Edition, 2005
R2	P.Radha Krishna, "Object Oriented programming through Java", CRC Press, 1st Edition, 2007
R3	S.Malhotra and S. Choudhary, "Programming in Java", Oxford University Press, 2nd Edition, 2014

Online Resources

Sl No	Website Link
1	http://java.sun.com 2. http://www.oracle.com/technetwork/java/index.html)
2	http://java.sun.com/javase
3	http://www.oracle.com/technetwork/java/javase/overview/index.html 5.
4	http://download.oracle.com/javase/7/docs/api/index.html
5	http://docs.oracle.com/javase/tutorial/
6	iiti.ac.in/people/~tanimad/JavaTheCompleteReference.pdf
7	https://www.codejava.net/books/4-best-free-java-e-books-for-beginners